Date: 25/8/25

Toc H Institute of Science & Technology

(An ISO 9001:2015 Certified Institution)

Arakkunnam

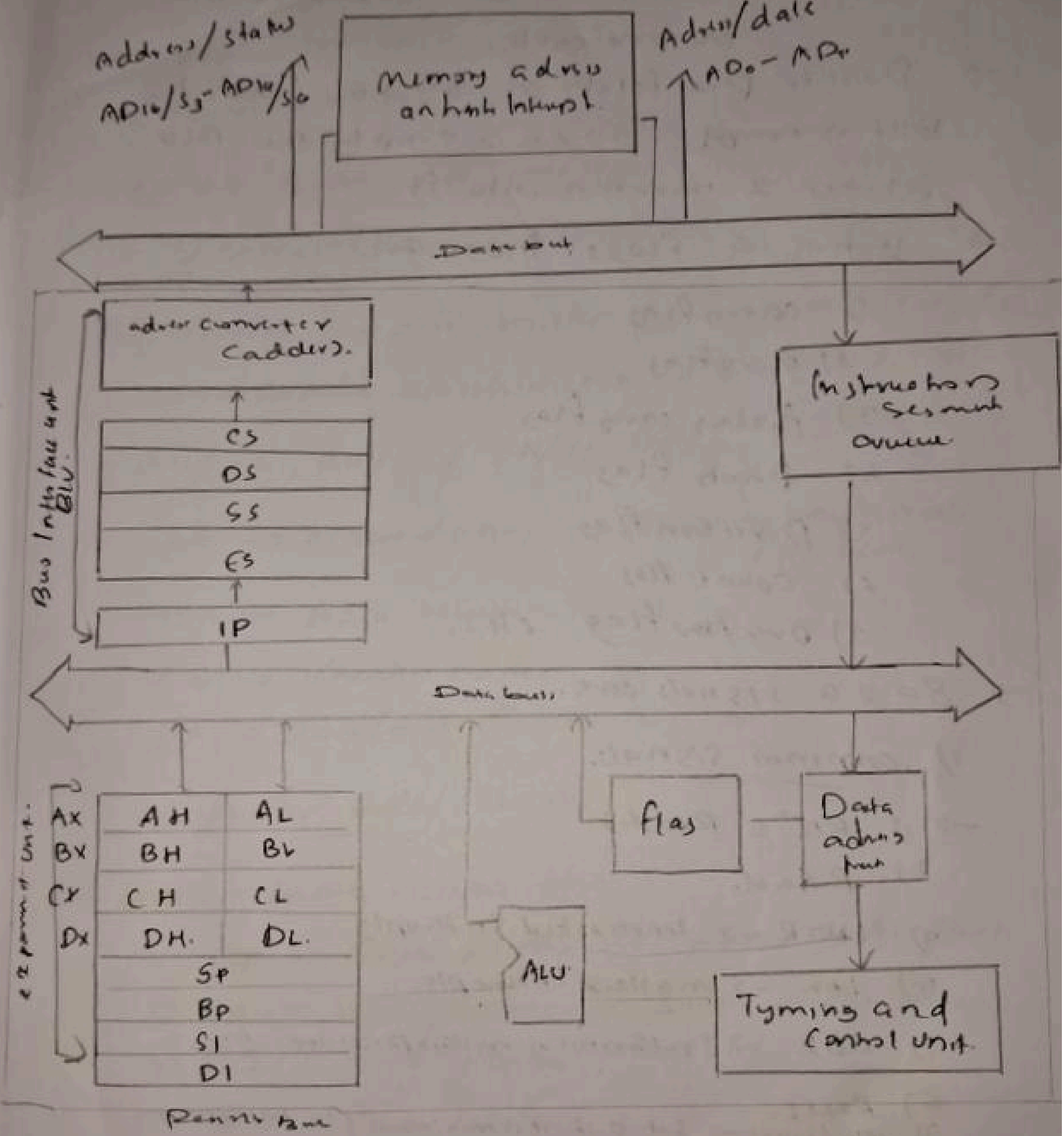
INTERNAL ASSESSMENT MAIN ANSWER BOOKLET

	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	XIII
a	3	3	1	0	3	10	6.	0					
b						4							
c													
d													
e													
f													
g													
h													
Total													
Grand Total	3	3	1		3	14	6						

TOTAL 30

Name	Dilsha P.P		
Branch & Semester	CSE - A S5		
Subject with Code	CST 307 Microprocessors and micro controllers		
Roll No.	49	University Reg. No:	T O C 2 3 C S 0 4 9.
Signature of Invigilator	[illegible]/8/25	No. of Additional Sheets	

6) a). 8086 Architecture



→ 8086 is a 16 bit micropocessor.

→ It has 16 bit adrss buss.

→ It is a pipelined architecture.

→ 8086 has memory banlars since it is

16 bit data bus.

→ Also it is faster as it can do 16 bit in one cycle.

→ There is pre fetch in 8086, when the ~~BIU is empt~~ 8080 is empty the BIU fetchs 2 numbers into it.

→ It has 9 flags. Those are.

1) carry flag
2) zeroflag
3) Auxilary carry flag
4) Parity flag
5) Direction flag
6) count flag
7) over flow flag etc...

→ 8086 signals are.

1) .common signals.

→ 1) RD' (Reads)
2) Read.
3) INTR → unmasked interupts
4) INW → masked Interupts.
5) CLK → Synchronizes microprocessor tim
6) Reset.
7) MIN/max: → Set 0 if it is minimum else maximum

2) Speecial minimum Signals

→ 1) $INT\bar{A}$
2) Lock
2) HOLD, HLDA

3) Special maximum signals.

1) status lines → S_0, S_1, S_2.
2) Queue → Q_1, Q_2 etc....

→ 8086 have (4 registers. Those are-

1) General purpose Register

The General purpose register are AX, BX, CX, DX and can be classified into 8 bit registers as,

→ AH/AL, BH/BL, CH/CL, DH/DL.

AX → Accumulator. → performs arithmetic operation,

BX → Base register.

CX → Counter register

10 DX → Data/address Register

2) Pointer Index Register

The pointer index registers are

1) SP → Stack pointer → points to the stack.
2) BP → Base pointer → points to the base
3) SI → Stack Interface
4) DI → Data Interface.

3) Segment registers

The segment resisters are

1) CS → Counter segment
2) DS → Data segment
3) SS → Stack segment

4) Extra segments → points to the extra value

→ The instruction sets in 8086 are.

1) Data transfer instruction

1) MOV Dest, src → copy contents of destination to source code.

es: MOV AX, BX.

2) PUSH src → push source into stack.

es: PUSH AX.

3) POP src → Pop source from stack into destination.

es: POP AX.

2) Arithmatic Instruction

1) ADD Dest, src → Addition of values

es: ADD AX, BX

2) MUL src → multiplication of values

es: MUL AX.

3) DIV AX → Division of values

es: DIV AX

4) IN AL/AX Port → Port from destination to port.

AL/AX 80H.

6) b).

8086	8088.
→ 16 bit microprocessor	→ 16 bit microprocessor.
→ 16 bit ~~address~~ data buss	→ 8 bit ~~address~~ data bus.
→ 9 flags	→ 9 flags.
→ It has memory banking since it has 16 bit ~~address~~ data buss.	→ It does not require memory banking since it has only 8 bit.
→ faster since it can complete 16 bit in one cycle	→ slower since it can complete only 8 bit in one cycle.
→ Supports pipelined architecture.	→ Supports pipelined architecture.
→ pipeline architecture is 6 bit	→ pipeline architecture is 4 bit.
→ Can prefetch data, as the BIU allows 2 number when it is empty.	can prefetch data, the BIU allows one number when it is empty.
→ ~~Address lines~~ It has $IO/\overline{M}$ to connect between memory and data...	→ It has $M/\overline{IO}$ to connect b/w memory and system.

8) CLC

MOV, SI 1000

MOV DI 2000

~~LDD~~ Mov AX.

mov AX (SI)

INC SI

INC SI.

7a) a) CMP → compare Instruction

→ CMP compare two numbers/obsile analyse and applies the condition.

eg: CMP AX, BX.

b) MUL → multiplication Instruction

→ MUL is the Multiplication Instruction is in arithmatic Instruction set.

→ It is an Instruction to multiply two numbers.

eg: MUL AX.

c) DIV → Division Instruction

→ DIV is the division operation that is in arithmatic instruction set.

→ It is an Instruction ~~to multiply~~ for two division of numbers

d) RCR → Right counter shift.

→ It refers arithmetic right shift.

eg: RCR 20

(d/e) ROR → Right Rotate Shift.

→ It ~~also~~ refers to arithmetic left and right shift.

eg: ROR 30.

3) Signals in maximum mode

1) S0, S1, S2 → status line.

2) Q0, Q1, Q → Ques.

explain others maximum mode signals also.

2) Pipelined architecture of 8080

→ 8086 supports pipeline architecture.

→ It is a 16 bit microprocessor.

→ 16 bit data buss

→ has 9 Flags.

→ It has memory banking since It has 16 bit data buss.

→ It is faster since It can complete 16 bytes in one cycle.

→ In the pipeline architecture of 8086, it has 6 ~~bits~~ bytes.

→ ... empty

BIU allows two numbers.

→ 8086 Architecture classified into Bus Interface Unit (BIU) and Execution Unit (EU).

→ BIU consists of Segment registers such as CS, DS, SS, ES.

→ EU consists of General purpose register and Pointer register.

5) i) Data transfer ~~registers~~ Instructions.

1) MOV Dest, src → moves from destination to source code.

eg: MOV AX, BX. (the value of BX is moved to AX)

2) XCHG src → exchange the values.

eg: XCHG AX.

3) PUSH src → push value to stack.

eg: PUSH AX.

4) POP src → pop values from stack to destination.

eg: POP AX.

2) Arithmetic Instructions

1) ADD Dest, src → Addition of two numbers and store result in left.

eg: ADD AX, BX.

Is fa... (EU)

MUL src → multiplication of a number ①

eg: MUL AX.

3) DIV src → Division of a ~~θ~~ number

eg: DIV AX.

4) IN AL/AX port → moving value to port.

eg: ~~AX~~ IN 80H.

5) OUT port AL/AX → moving value from port to destination

3

eg: 80H OUT.

---

2) In 8085 Architecture it has following parts.

1) Interrupt I/O control → control Interrupt using signals INTR, RST 5.5...
2) serial I/O control → control serial communication by signals SID, SOD.
3) Accumulator → performs arithmatic operations and store result (Helps)
4) temporary register → store a value during arithmatic operation
5) Instruction register → ... instruction.
6) multiplexer array
7) Register array
8) stack pointer → points to stack.
9) program counter → execute next instruct.
10) time and control unit etc....
11) ALU → performs Arithmatic operations

3

8

4) i) MOV [AX], [BX].

CLC

⇒ MOV 1000, 2000

ii) MOV AX, 5000 [BX]

→ CLC.

MOV 1000, 5000 2000